Solar Energy & Energy Transition

First Solar (NASDAQ: FSLR)

FSLR is America's leading photovoltaic (PV) solar technology manufacturing company, and the only U.S. headquartered company among the world's largest solar manufacturers. Unlike its foreign competitors, FSLR's modules contain 0% polysilicon and primarily consist of cadmium telluride (CdTe), reducing exposure to harsh tariffs and Foreign Entity of Concern (FEOC) classifications. We rate FSLR a Buy with a 12-month base-case target of $234.72, representing 36.9% upside on its November 13, 2026 closing price.

Investment summary

FSLR is the one large-scale U.S. module maker which has minimal reliance on Chinese trade. Because their modules don't contain any polysilicon, FSLR doesn't engage in trade with Xinjiang, which manufactures 35 – 45% of the world's polysilicon. This allows FSLR's costs to be minimally impacted by Section 232, Section 301, and FEOC rules while similar-scale competitors are heavily affected.

Incentive programs provide multi-year earnings tailwinds through 2029. In FSLR's two most relevant markets, the United States and India, government production credits provide FSLR with per watt credits for simply manufacturing solar modules in domestic American and Indian facilities.

Prices of solar modules are expected to rise. From December 4, 2026, onwards, solar module imports must be priced at a minimum of $0.38/W and $0.22/W for cells plus a 15% tariff. For U.S. module manufacturers who import solar cells, Roth Capital Partners estimates that post-232 pricing may rise to $0.40/W. FSLR makes its own thin-film modules in the U.S., so their asking price can rise while costs stay flat.

We rate FSLR a Buy with a 12-month target of $234.72 and 36.9% implied upside. We used a forward EV/EBITDA multiple for our price target, with DCF valuation and comparable analysis as cross checks.

PELAGIC LENS

FSLR has a strong commitment to "Responsible Solar." Their modules are manufactured through a vertically integrated process that uses less energy and water than conventional polysilicon modules.

RATING
Buy

12-MONTH TARGET
$234.72

PRICE (AS OF 11/13/2026)
$171.39

IMPLIED UPSIDE
36.9%

BEAR / BULL
$156.00 / $280.21

MARKET CAP
$18.42B

SHARES OUTSTANDING
107.5M

ENTERPRISE VALUE
$17.12B

52-WEEK RANGE
$165.90 – $320.95

FY26 VOLUME GUIDE
16.0 – 17.0 GW

BACKLOG
44.0 GW

Business Model

How does First Solar make money?

22.1 GW	44.0 GW	$2.4 – 2.6B
Nameplate capacity	Contracted backlog	FY2026 gross profit guidance

Company Overview

FSLR, founded in 1999, is a leading vertically integrated PV solar technology manufacturing company, and the only U.S. headquartered company among the world's largest solar manufacturers. They operate mostly in the Western hemisphere and have manufacturing facilities in Ohio, Alabama, Louisiana, and South Carolina, alongside international facilities in Malaysia, Vietnam, and India.

Customers

FSLR sells mainly to utility scale developers, power producers, and commercial buyers, and about 96% of its sales are to the United States. Customers commit on purchases years ahead: at September 30, 2026 the contracted backlog was 44.0 GW, worth about $13.5B. Revenue follows two factors, gigawatts sold and the price per watt.

Exhibit 1: Core Products

Product	Description
Series 6 Plus	4x6 ft older module averaging 464 W, mostly made in international plants, but production cut due to weak pricing
Series 7	4x7 ft module, current main product, averages about 532 W, made mainly in the U.S. (qualifies for the full $0.17/W credit) and India
Series 7 CuRe	Upgrade to the Series 7 that replaces copper in the semiconductor with more stable Group V elements. Currently only permanently produced in Ohio, and planned to be rolled out across other plants in phases
Perovskite (R&D)	Next-generation technology in research and development, 1 GW pilot line targeted for operational readiness by H1 2027, but not yet commercially available

Sources: Perovskite Info, FY2025 10-K

Where does profit come from?

A fully US-made module costs about $0.267/W to make and historically sold for $0.31/W in the backlog, not including technology adjusters. Pre-credit margins are thin, being only 4 cents, while the 45X credit adds up to 17 cents. That is why credits were $1.5B, or roughly 82% of gross profit, in the first nine months of 2026. It also explains why management cut 2026 sales and volume guidance in their Q3 10-Q but left gross profit guidance untouched: the lost volume was mostly international where no credits are earned.

Policy Moat

Why does First Solar have a policy advantage, and how durable is it?

45X Manufacturing Credits

Former President Joe Biden's Inflation Reduction Act (IRA) of 2022 created the Advanced Manufacturing Production Credit under Section 45X of the U.S. Internal Revenue Code. To put it simply, 45X is a federal tax credit for clean energy components made in the United States, and for solar it pays by the watt. A fully U.S. made module earns up to $0.17/W, and a module finished in the U.S. from imported cells pays about $0.07/W. FSLR books the credit as a reduction in cost of sales and has begun transferring credits to cash, so the benefit hits both earnings and cashflow.

Which policies matter, and until when?

Four pieces of federal policy shape FSLR's advantage, and not all of them are in permanent effect:

Exhibit 2: The policy stack

Policy	What it does	Impact on FSLR	Status and date
45X Credit	Pays a federal credit on U.S. made modules, up to $0.17/W	About $2.1B in FY2026 guidance, around 82% of 9-month gross profit	Full value through 2029, then 75% in 2030, 50% in 2031, 25% in 2032, and 0% 2033 onwards
48E and 45Y Project Credits	Tax credits for developers building solar projects. Over 40% of the solar project's equipment cost must be from non-FEOC sources	Supports demands for solar modules	Projects had to begin construction July 4, 2026, or be in service by YE2027
FEOC Rules	Restrict incentives for projects tied to China, Russia, Iran, and North Korea	An edge over China-linked polysilicon makers	Notice 2026-15, February 12, 2026
Section 232	Sets minimum import prices of $0.38/W for modules and $0.22/W for cells, plus a 15% tariff	Raises import costs and supports domestic pricing	Effective December 4, 2026, and rates can differ by country

Sources: FSLR Q3 2026 10-Q, 26 U.S.C. §45X, IRS Notice 2025-42, Notice 2026-15, White & Case

Is the moat permanent?

No, and the schedule is the main concern for FSLR. The credit holds at full value for modules sold through 2029, then steps down to 75% in 2030, 50% in 2031, and 25% in 2032. It ends in 2033 assuming no extension, so FSLR's advantage narrows from 2030 onwards. The 48E and 45Y project credits work in a similar sense. Because projects had to begin construction by July 4, 2026, to receive the credit, demand after these projects are completed in 2028 – 2029 is uncertain. FEOC Rules and the Section 232 floor work in FSLR's favor and are not on the same timer. While legislation could change this, they both don't currently have hard deadlines that FSLR has to work around.

Competition and Capacity

Who can compete with First Solar in the U.S.?

65 GW
U.S. module assembly capacity

3.2 GW
U.S. cell capacity

$0.22/W
Section 232 minimum cell import price

The silicon supply chain handicap

About 65 GW of U.S. module capacity sits on only 3.2 GW of U.S. cell capacity, so most silicon assemblers import their cells from Asia. From December 4, 2026, onwards, Section 232 sets a $0.22/W minimum import price on those cells. Furthermore, a module finished in the U.S. from imported cells earns only about $0.07/W of 45X credit against $0.17/W for a fully U.S. made one. FSLR's thin film modules are unique as they do not use traditional individual solar cells. Instead, they manufacture thin-film photovoltaic (PV) modules. This process is much faster than polysilicon module manufacturing, and it uses 2 to 3% as much semiconductor material as polysilicon.

Exhibit 3: Silicon Makers vs. First Solar

Metric	Silicon Makers	First Solar
Manufacturing process	Several days and through multiple factories	Hours, one continuous line
Semiconductor material	Baseline	2 – 3% as much as baseline
Reliance on China	Polysilicon, wafers, cells	Tellurium
45X credit per watt	About $0.07/W if cells were imported	$0.17/W for fully U.S. made modules

Sources: FSLR FY2025 10-K, Q3 2026 10-Q, 26 U.S.C. 45X

Rival ownership and credit exposure

FSLR's FY2025 10-K names JinkoSolar, JA Solar, Trina, Canadian Solar, and Hanwha Qcells as rivals, and FEOC rules are splitting that group. JinkoSolar is likely to fail the non-FEOC ownership test, which puts the credits tied to U.S. products at risk. Canadian Solar has not completed a sell-off of its Chinese-linked ownership, so its eligibility to receive manufacturing credits is also uncertain until it does. Trina and JA Solar have already sold their U.S. plants, which reduces their exposure by leaving U.S. manufacturing. First Solar has also sued JinkoSolar and Canadian Solar over TOPCon patents, but a decision is still pending.

How does First Solar defend its lead?

FSLR has 22.1 GW of nameplate capacity and sold 17.5 GW of modules in 2025. Management expects about 80% of that capacity to be in the U.S. by 2027, which raises the share of its output that qualifies for the 45X credit. It is also enforcing its patents: it sued JinkoSolar and Canadian Solar over TOPCon and, on February 24, 2026, filed a U.S. International Trade Commission (USITC) petition naming 10 companies. FSLR's advantage over competitors is real and strong but policy dependent.

Pricing and Backlog

Is First Solar's backlog enough?

44.0 GW
Contracted backlog, about $13.5B

$0.31/W
Average backlog price

$0.36/W
Latest U.S. bookings

What is in the backlog?

FSLR's backlog is the total value or volume of confirmed customer orders that it has received but has not yet fulfilled or delivered. The backlog was 44.0 GW at September 30, 2026, worth about $13.5B, and management says it extends through 2030. It was 45.1 GW at the end of Q2 2026, so it fell about 1.1 GW while the company shipped 4.1 GW total, which implies roughly 3 GW of net new bookings. It also shrank over a longer stretch, from 78.3 GW at the end of 2023 to 50.1 GW at the end of 2025.

Are new orders priced better?

Yes, at least in the U.S., newer orders are expected to be priced better due to FEOC rules and the Section 232 tariffs:

Exhibit 4: The price ladder

Price point	Price	Note
Backlog average	$0.31/W	$13.5B across 44.0 GW
India bookings, year to date	$0.20/W	1.3 GW
U.S. bookings since July 30, 2026	$0.36/W	1.4 GW
Pre-Section 232 import price	$0.27/W	Before December 4, 2026
Post-Section 232 import floor	$0.44/W	$0.38/W plus a 15% tariff
Roth estimate for U.S. assemblers after Section 232	$0.40/W	29% higher price than backlog average

Sources: FSLR Q3 2026 10-Q, FSLR Q3 2026 earnings release, Taiyang News, Roth Capital Partners

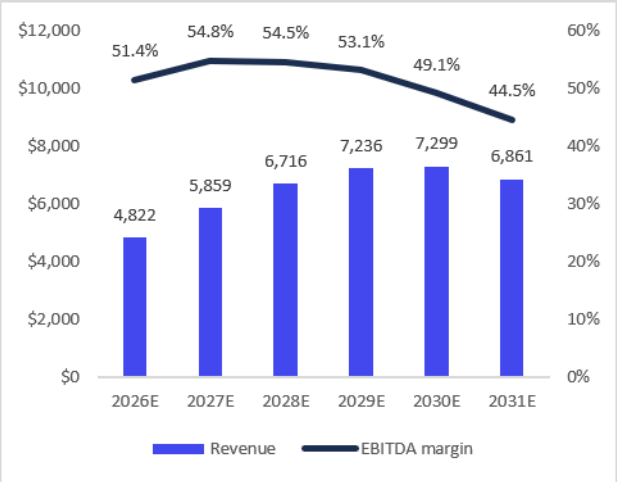
Why might orders lag the price increase?

Management cut 2026 sales guidance to $4.6B – $4.9B from $4.9B – $5.2B and volume guidance to 16.0 – 17.0 GW from 17.0 – 18.2 GW, citing slower orders and tariff uncertainty. Although most of the volume guidance cut was international, where the 45X credit isn't applied, the decrease may still be seen as a cause for concern. A Roth Capital Partners analyst said higher module prices after Section 232 are likely to benefit FSLR, but may take time to turn into actual bookings, with more clarity coming in the first half of 2027. Cancellations are the other risk. In 2025, gross bookings were 7.4 GW against 8.3 GW of cancellations, so net bookings were actually negative.

Financials and Earnings Quality

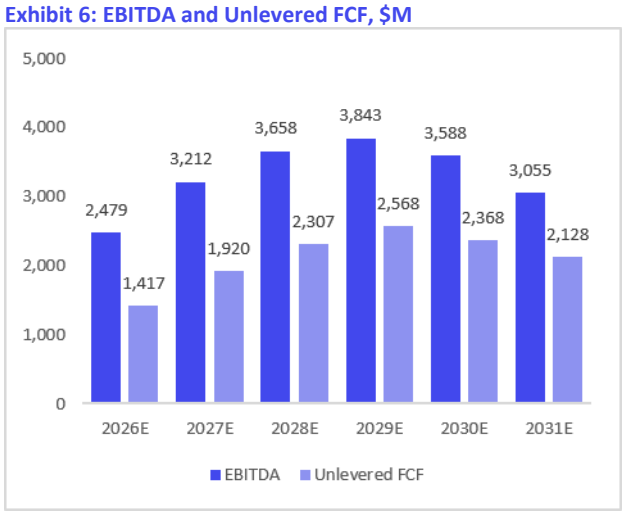
Are First Solar's earnings real, and how much of them depend on the 45X credit?

Exhibit 5: Revenue and EBITDA margin, $M



Sources: Pelagic estimates, S&P Capital IQ

Exhibit 6: EBITDA and Unlevered FCF, $M



Sources: Pelagic estimates, S&P Capital IQ

Exhibit 7: Financial summary

$M	FY2024A	FY2025A	FY2026E	FY2027E
Revenue	4,206.3	5,219.4	4,822.0	5,858.7
EBITDA	1,817.2	2,126.1	2,478.8	3,212.3
EBITDA margin	43.2%	40.7%	51.4%	54.8%
Net income	1,292.0	1,528.2	1,817.7	2,369.6
EPS ($)	12.02	14.21	16.91	22.04
Capital expenditure	1,526.1	869.9	730.9	890.5
Unlevered free cash flow	25.0	1,508.1	1,416.9	1,920.3
Net debt / EBITDA	(0.7x)	(1.1x)	(0.7x)	(1.1x)

Sources: FSLR FY2024 10-K, FSLR FY2025 10-K, S&P Capital IQ, Pelagic Estimates

Financial summary

FSLR's earnings are set to grow faster than its revenue. We model revenue falling 8% to $4.82B in 2026 as volume dips from 2025, yet net income still rises 19% to about $1.82B because of a widening EBITDA margin. In 2027, we project that revenue will grow 21% to $5.86B on higher-priced bookings, causing an increase in net income. Margin is the main driver of earnings for FSLR, so the quality of earnings depends on what supports that margin.

What's driving the margin?

The main driver of FSLR's stellar margins is the 45X credit. It is booked as a reduction in cost of sales, so it sits inside gross profit and EBITDA rather than below the line. FY2026 guidance implies about $2.1B to $2.2B of credits, with roughly $1.5B recognized in the first nine months, about 82% of gross profit over that period. FSLR has also begun transferring credits to cash, so the effect on cashflows is also tangible. The main risk with these credits is that they will fade over time. Assuming no extension, the credit steps down to 75% in 2030, 50% in 2031, and 25% in 2032, and ends altogether in 2033. Our model has EBITDA margin peaking at 54.8% in 2027, and falling to 44.5% by 2031, and our DCF valuation removes $1.05B of credits from the terminal value, so it does not assume that the credit lasts forever. Without the credit, margins depend on the difference between the post-credit module price and a production cost of $0.267/W.

Capital Expenditures and Debt

Capex is FSLR's largest draw on cash, but it fell from $1.53B in 2024 to $0.87B in 2025 as their factory build-out slowed. We predict roughly 15% of sales to go to capex in 2026 and 2027, $0.73B and $0.89B respectively. With that, unlevered free cash flow recovers from around break-even in 2024 to $1.51B in 2025, $1.42B in 2026E and $1.92B in 2027E, which is 57% to 60% of EBITDA. Working capital absorbs about $250M a year in our forecast, which is why the conversion rate isn't higher. FSLR currently has zero long term debt and a net cash balance of $1.5B, so the company can fund any planned capex and its $500M buyback authorization without excessive borrowing.

Valuation

What is First Solar worth in 12 months?

Methodology

We value FSLR on NTM EV/EBITDA. In November 2027, the next twelve months will be mostly 2028, so we apply a multiple of 5.94x to our 2028E EBITDA of $3,658M. That is FSLR's average NTM multiple in November against 5.77x at the November 13 close after the post-midterm pullback. We assume it recovers because we see no change in the company's earnings or guidance. That gives an enterprise value of $21.7B. We then add $3.5B of net cash to reach our equity value, because enterprise value excludes cash. The $3.5B is the $1.5B reported at September 30, 2026, plus about $2B of free cash flow after capex, with no buybacks assumed. Dividing by 107.5M shares gives a share price of $234.72.

In November, FSLR traded at an average of 5.95x NTM EV/EBITDA against a peer median of 11.1x. Peers range from Array Technologies at 7.38x to Enphase at 17.97x. Applying the two lowest peer multiples, Array (7.38x) and Canadian Solar (9.24x) to FSLR's current NTM EBITDA gives about $218 to $269 a share respectively. The main risk to the target is the fact that the 45X credit isn't permanent, but we believe that the phase down period is far enough away that the market will be mainly focused on FSLR's current financial performance.

A discounted cash flow model acts as our cross-check. It values the stock at $177.47 using an 8.0x exit multiple (WACC of 12.24%) and $156.83 using the perpetuity growth method, close to the stock's current price. The DCF removes $1.05B of 2031 credit from the terminal year, so it prices the post-credit business, while our target multiple is applied to earnings that include the credit.

Exhibit 8: Scenario analysis

Metric	Bear	Base	Bull
2028E EBITDA	$2,653M	$3,658M	$3,869M
Valuation multiple	5.0x	5.94x	6.88x
Target price	$156.00	$234.72	$280.21
Upside / downside	(9.0%)	36.9%	63.5%

Sources: Pelagic Estimates, S&P Capital IQ

Key Assumptions

Three inputs drive our target price: 2028E EBITDA, the valuation multiple, and net cash. Our base case assumes that revenue growth of 21.5% in 2027E fades to -6% by 2031E and an EBIT margin of 43% falls to 30.6%. Our bear case is more conservative: revenue growth of 15% in 2027E descends to -10% in 2031E. The EBIT margin starts off smaller, at 35%, and lands at 25.5% by 2031E. We also apply a smaller valuation multiple of 5.0x, 0.94x lower than our base case. Finally, our bull case is the most optimistic. Revenue growth of 26.7% in 2027E is high, and our EBIT margin starts slightly higher at 43.5% and lands at 35%. Net cash is the smallest driver, and we leave it constant throughout our three cases: $1.5B plus about $2B of free cash flow.

Exhibit 9: Discounted cash flow analysis, $M

$M	2026E	2027E	2028E	2029E	2030E	2031E
Revenue	4,822.0	5,858.7	6,715.6	7,236.0	7,299.4	6,861.4
EBITDA	2,478.8	3,212.3	3,658.0	3,843.1	3,587.6	3,054.7
EBIT	1,898.7	2,507.5	2,894.4	3,068.1	2,854.0	2,410.4
Unlevered FCF	1,416.9	1,920.3	2,306.5	2,567.7	2,368.0	2,127.5

Metric	$M
PV of free cash flow	8,581.7
PV of terminal value	8,868.8
Enterprise value	17,450.5
+Cash	1,624.1
Equity Value	19,074.6
÷ Shares	107.5
DCF value per share	$177.47

Sources: Pelagic Estimates, S&P Capital IQ

Exhibit 10: Comparable company analysis

Company	Market Cap ($M)	EV ($M)	NTM EV/EBITDA	LTM EV/EBITDA	NTM revenue growth	NTM EBITDA margin
First Solar	18,421.0	17,122.0	5.77x	6.82x	19.98%	56.22%
Canadian Solar	670.0	8,000.5	9.24x	12.4x	26.35%	15.57%
JinkoSolar	492.3	6,515.3	11.68x	11.9x	42.86%	5.28%
Nextpower	14,309.2	13,323.8	12.21x	15.41x	22.68%	21.25%
Array Technologies	746.7	1,872.7	7.38x	19.2x	13.95%	15.58%
Enphase Energy	3,449.4	3,132.1	17.97x	33.1x	NM	20.28%
Shoals	1,396.6	1,589.1	10.47x	13.6x	8.2%	21.9%
Median of Peers	--	--	11.08x	14.51x	22.68%	17.93%
Mean of Peers	--	--	11.49x	17.60x	22.81%	16.64%

Source: S&P Capital IQ

Risks

What could damage the thesis?

Post-credit pricing and margins (High)

The 45X credit steps down to 75% in 2030, 50% in 2031, and 25% in 2032, before vanishing altogether in 2033. The backlog is priced at $0.31/W, and against a production cost of $0.267/W, that leaves an ex-credit margin of only $0.04/W. Pricing today is supported by the Section 232 import floor starting December 4, 2026, so if new bookings came in below the recent $0.36/W level, that could put the ex-credit margin at risk. Our model already has EBITDA margin falling from 54.8% in 2027E to 44.5% in 2031E. If it falls faster, both 2028E EBITDA and the multiple would suffer, making the bear case more likely.

Demand after developer credits end (Medium-High)

New solar projects may find it hard to qualify for investment and production tax credits. The July 4, 2026, deadline has already passed, so projects must be in service by YE2027 to qualify. The contracted backlog of 44.0 GW extends through 2030, but the company has not said how much is scheduled for each year. Backlog fell by 1.1 GW last quarter while 4.1 GW were shipped, which implies about 3 GW of net new orders. If new orders keep running below shipments, volume after 2028 could fall faster than our forecast, which would lead to a lower EBITDA and share price.

Multiple compression (Medium)

Our price target holds the multiple near 5.94x, the November average, against 5.77x at the November 13 close. Every 1.0x above or below is worth about $34 a share. If it falls to our 5.0x bear case, base-case EBITDA still gives about $203, 18% above the current price, so the upside would shrink but not disappear. The multiple could also fall further as the 2030 phase-down approaches.

Policy and trade (Low)

The 45X credit was about 82% of gross profit in the first nine months of 2026, so any change to it, to the FEOC rules that govern which customers claim credits, or to the Section 232 import floor would hit FSLR's earnings directly. Even though the effect would be large, the Democrats recently swept Congress during the 2026 midterm elections, so the likelihood of such a change is low. A weaker import floor would directly feed into the first risk.

Sources

[1] FSLR 10-K, FY2025
[2] First Solar Q2 2026 earnings release (8-K)
[3] First Solar Q2 2026 10-Q
[4] White & Case on the Section 232 action
[5] 26 U.S.C. §45X
[6] India awards financial incentives to First Solar's Tamil Nadu manufacturing facility
[7] PLI (Production-Linked Incentive)
[8] US Section 301 Tariffs Split ASEAN Exporters in 2026
[9] In a first, First Solar cashes out 45X credits
[10] Trump signs Section 232 tariffs, placing minimum import price on polysilicon imports
[11] First Solar Q3 2026 10-Q
[12] FSLR FY2024 and FY2025 Financials
[13] FSLR Q4 2025 Results
[14] IRS Notice 2025-42
[15] RSM, beginning-of-construction rules for solar and wind
[16] FSLR authorizes $500M buyback
[17] FSLR warns of order cliff by 2028
[18] Taiyang News
[19] PV Tech
[20] PV Magazine
[21] Perovskite Info
[22] FSLR FY2024 10-K
[23] Foley & Lardner LLP
[24] S&P Capital IQ

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